DPS hardware for vertical drift

Existing BDE design, additions and procurement

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I&I workshop draft — 27 October 2026

DEEP UNDERGROUND NEUTRINO EXPERIMENT

Fermilab

Use a commercial PLC for VD detector protection

Proposal: use the bottom drift electronics (BDE) design and add the inputs and outputs needed by the other VD systems.

Put the central programmable logic controller (PLC) in the CUC, with distributed I/O near the equipment.

Use OPC UA to connect DPS status and agreed commands to slow controls.

CUC = Central Utility Cavern; central-PLC location proposed. HD can later use an extension or a separate PLC. This plan covers VD.

The VD rack plan already reserves space for DPS

Rack type	Count	DPS equipment / connections
BDE auxiliary	10	12U per rack; eight WIEC connections per rack
BDE low voltage	20	One PL506 per rack; four WIECs per supply
Bias high voltage	10	12U per rack for the DPS interface to ISEG supplies

Use these locations and rack allocations to place the additional I/O.

May 2025 VD rack workbook; June 2026 BDE layout. U is rack height. These are design allocations; the bias-HV sheet is a first estimate.

One BDE I/O group serves eight WIECs

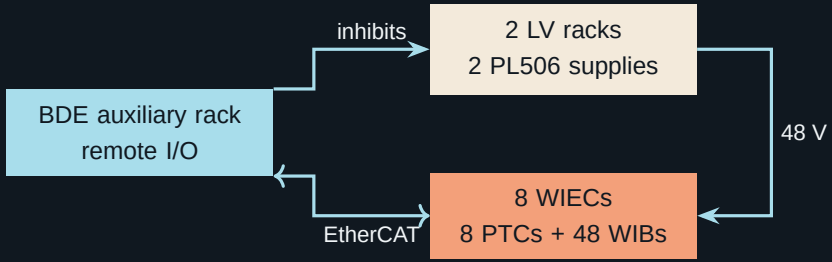


Diagram. Ten groups: 20 PL506 supplies, 80 WIECs and 480 WIBs.

The auxiliary rack also connects the fans, heaters and temperature sensors.

WIEC = warm interface electronics crate; PTC = power and timing card; WIB = warm interface board. June 2026 layout; fan/heater wiring omitted here.

Panel drawings and the Ignition front end are available

TPC protection panel designs

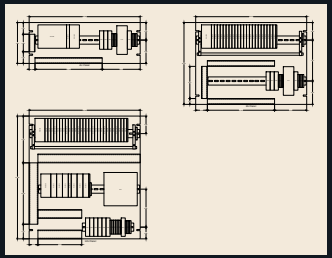
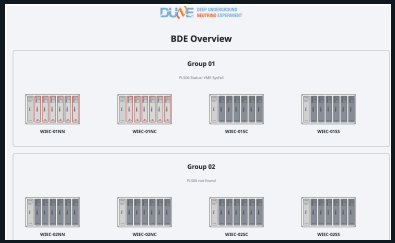


Diagram. Central PLC, bias I/O and LV/fan/heater panels.

BDE Ignition screens



Existing display of WIEC groups; June 2026 SC progress presentation.

Parts list: CX5140 PLC, EK1100 couplers, EK1521 fibre links and EL2124 inhibit outputs.

TPC reference: EDMS 2401090 v4, sheet 2. Use the VD rack allocations when adapting these panels.

The VD bias-panel draft specifies two DIN rails per rack

Component	Per rail	10 racks
EK1100 EtherCAT coupler	1	20
EL9410 supply terminal	1	20
EL2124: four 5 V outputs	18	360
EL9011 end cap	1	20
24 V power supply	1	20

Each rail occupies 337 mm. The draft provides 1,440 output channels; the working VD inventory lists 576 bias channels.

May 2025 workbook, Bias HV Rack!K17:K21. First-pass quantities: reconcile the inhibit allocation before purchasing.

VD rack power: load estimates and supply ratings

Equipment	Per rack	Basis in the workbook
PL506 in 20 BDE LV racks	2.254 kW input	Four WIECs; 75% supply efficiency assumed
DPS in 10 BDE auxiliary racks	831 W estimate	Six-WIEC source calculation; note calls for about 33% more for VD
DPS in 10 bias-HV racks	960 W listed	Equals the 24 V, 40 A supply rating; channel load is not calculated here

Recalculate the eight-WIEC auxiliary load and the bias-panel load before sizing the added equipment.

May 2025 workbook: BDE LV RACK!D27,K27:K29; BDE Aux Rack!D14,F14,K13:K19; Bias HV Rack!D15,F15,K22.

The PX3 rack-protection setup already defines shutdown

June 2026 setup guide	Documented behaviour
Ignition connected	Ignition reads sensors and switches outlets over SNMP
Ignition unreachable	PX3 local rules handle smoke, water and temperature; the network-switch outlet stays on
After a fallback trip	Power is restored manually or through Ignition

Agree where the VD rack-protection logic will run: PDU, DPS PLC and Ignition each need a defined role.

DUNE-doc-32655 v8, PX3 Rack Protection Setup, pp. 3–6 and 14–16. Confirm the selected rack/PDU configuration against this reference setup.

Use the inventory to specify the additional I/O

Existing material	Work for VD
Rack and panel inventory	Allocate terminals, power and cable routes for added signals
Subsystem hardware lists	Remove overlaps; complete CalCI and other missing connections
Interface documents and matrix	Resolve conflicting actions; record signal type, timing and feedback

Example: 80 PoF boxes × (8 temperatures + 8 voltages) = 1,280 monitoring values.

Monitoring values are not automatically PLC terminals. Choose the readout and protection interfaces before counting modules. CalCI = calibration and cryogenic instrumentation; PoF = power over fibre.

Confirm the time allowed for each protective action

So far, I have not found a critical protection function that needs a response below 1 ms.

Each consortium must specify the time from a hazardous condition to the completed protective action.

Include detection, communication, PLC processing and equipment shutdown.

Use the simulation framework to prepare tests. Measure complete paths on hardware; add faster local protection wherever a confirmed requirement needs it.

Resolve the interfaces that affect hardware

Interface	Decision needed
SC / DPS	With Geoff, agree command priority, reset and re-enable; use SC for relevant long-term records
Cryogenic measurements	Confirm each source and whether DPS needs a dedicated signal or a software exchange
LBNF/DUNE and SDSTA BMS	Select the limited hardware signals and their connection point; define software monitoring
CUC / detector mezzanine	Specify fieldbus routes, networks and control-power supplies

BMS = building management system. Keep the two BMSs distinct. SDSTA retains responsibility for personnel-safety alarms.

Use the SURF inspection to map the facility connections

1

Already prepared

Marked maps for levels 4850 and 4910, with drawing references and a panel survey register.

2

Record on site

Controller and I/O models, panel labels, available terminals and connection routes underground and at Ross Dry.

Coordinate access with Rachel; ask Dale Curran about the SDSTA interfaces.

Inspection planned for this week, subject to access and safety arrangements. Maps are prepared but have not yet been shared.

Use the existing test stands for the VD additions

Test	Measure or demonstrate
PLC, remote I/O, PTC and supply inhibit	Complete trip time and equipment feedback
PDU sensors and shutdown logic	Response with SC connected, lost or stalled
SC commands, reset and re-enable	Protection takes priority; recovery is deliberate
Fieldbus, network and power failures	Output states, lost branches and restart behaviour

Fermilab BDE/VST and G271 material already exists. Agree test time, configuration and extra equipment with the owners; identify any SURF test needed.

Prepare the VD design and purchases for June 2027

Step	Result
Reconcile existing records	One VD rack, panel and I/O schedule; list added signals and unresolved requirements
Design and prototype	Updated drawings, power budget and test results for the added interfaces
June 2027 review	Parts list, costs, lead times, spares, installation sequence and procurement readiness
After review	Order approved hardware; install against confirmed site dates

Check possible summer 2027 facility-interface needs now. Full VD response-matrix review targets early 2028; hardware requirements are needed before purchasing.

Agree who does the work

Proposed team	Next action
Manuel + control engineer	Specify VD additions, cost and procurement work
Terri, Trevor + Manuel	Reuse the BDE hardware design and test arrangements
Manuel + consortia	Resolve missing signals, timing and conflicting requirements
Manuel + Geoff	Agree SC/DPS commands, rack protection and recovery
Vivian + subsystem / DPS teams	Update ICDs and record the agreed interfaces

Confirm support and dates at I&I. Agree who protects equipment in each installation and operating phase before DPS is available.

Reference: the May 2025 VD rack inventory

Rack type	Count	Rack type	Count
BDE low voltage	20	PDS low voltage	4
BDE auxiliary	10	Drift HV	2
TDE	20	Reserved drift HV	2
Bias HV	10	Spare / CalCI	8

76 numbered mezzanine positions: SEDM-01 to SEDM-76.

The worksheet also has a separate concept for 40 top-plate mini-racks.

Counts are rack positions, not numbers of PLC panels or remote-I/O stations. Reconcile this May allocation with later changes before ordering.